

```

@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);
}
}

```

▪ Comparison :

Feature	Java (Android)	Flutter (Dart)
UI Structure	Uses separate XML files for layout	UI is built directly in Dart code
Navigation Setup	Requires explicit Intent or Fragment management	Handled easily using MaterialApp
Complexity	More verbose and requires manual UI updates	More streamlined with a declarative UI model

Step 9 : What is Scaffold in Flutter?

a. What is Scaffold?

- Scaffold provides the basic structure or skeleton for a Flutter app screen.
- It organizes key UI elements like the AppBar, body, and FloatingActionButton.
- It simplifies UI layout, making it easier to structure an app's interface.
- Key UI Components Inside Scaffold:
 - AppBar → The top bar with a title or navigation buttons.
 - body → The main content area of the screen.
 - FloatingActionButton → A floating button for quick actions.

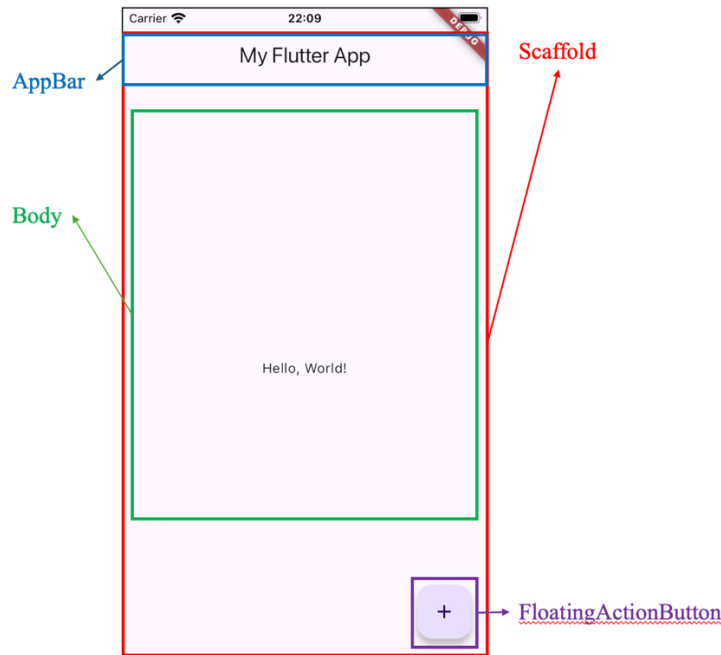
b. Basic Example of Scaffold

- Flutter Code Example :

```

Scaffold(
  appBar: AppBar(
    title: Text('My Flutter App'),
  ),
  body: Center(
    child: Text('Hello, World!'),
  ),
  floatingActionButton: FloatingActionButton(
    onPressed: () {
      print('Button Pressed!');
    },
    child: Icon(Icons.add),
  ),
);

```



- **Key Parts of Scaffold**
 - ◇ AppBar - The Top Navigation Bar
 - Creates a top bar for navigation and titles.
 - Often contains text (title) and action buttons.
 - Uses the AppBar widget inside the appBar property.
 - ◇ body - The Main Content Area
 - The largest section where most widgets (texts, images, buttons) are placed.
 - Requires a widget inside, like Center, Column, or Row, to organize content.
 - Often contains other widgets like Text, ListView, GridView, etc.
 - ◇ floatingActionButton - Quick Action Button
 - A button that “floats” over the content, often used for quick actions.
 - Requires an icon or another widget inside.
 - Commonly used for actions like adding items or refreshing content.
- c. **Why is Scaffold Important?**
 - Provides a structured layout, making it easier to build screens.
 - Ensures a consistent UI experience across different parts of the app.
 - Integrates well with MaterialApp, enabling smooth navigation and theming.
 - **Key Benefits of Scaffold:**
 - *Provides a well-defined structure* → Simplifies UI design.
 - *Enhances user experience* → Ensures a modern and responsive layout.
 - *Integrates with Material Design* → Follows best practices for UI consistency.

Summary of Flutter and Dart Project Code Structure

- Flutter apps start from `main()` and `runApp()`, which load the `MyApp` widget.
- `MyApp` extends `StatelessWidget` and returns a `MaterialApp`, the root widget.
- Inside `MaterialApp`, `Scaffold` is used to structure the screen layout.
- The `build()` method constructs the widget tree, and return specifies the UI.
- Key widgets like `AppBar`, `body`, and `floatingActionButton` are placed inside `Scaffold`.